

Emily Wong

Email: emily.wong@systems.com

Phone: +1 (703) 333-4444

Location: Reston, VA

LinkedIn: [linkedin.com/in/emilywong-backend](https://www.linkedin.com/in/emilywong-backend)

PROFESSIONAL SUMMARY

Senior Backend Engineer with 8 years of experience building high-availability, low-latency microservices and REST APIs. Expert in Java/Spring Boot and proficient in integrating these services with large data infrastructure like Kafka and SQL databases. Experienced in high-scale cloud deployment using AWS.

EDUCATION

Master of Science in Software Engineering

Virginia Tech, 2016

Bachelor of Science in Electrical Engineering

University of Maryland, 2014

CERTIFICATIONS

- Oracle Certified Professional, Java SE 11 Developer
- AWS Certified Solutions Architect - Associate

WORK EXPERIENCE

Principal Backend Engineer | Capital One | January 2020 - Present

- Designed and developed critical financial microservices in Java/Spring Boot, handling over 100M daily transactions with 99.99% uptime.

- **Implemented Kafka producers and consumers** to integrate financial transactions into the central data lake, ensuring exactly-once processing semantics.
- Led the migration of monolith services to a containerized architecture using Docker and Kubernetes on AWS ECS.
- Optimized PostgreSQL database performance, reducing API response times by 40% across key endpoints.

Software Engineer | Lockheed Martin | June 2016 - December 2019

- Developed high-performance C++ and Java applications for defense systems.
- Managed and queried large, complex SQL databases (Oracle).

TECHNICAL SKILLS

Programming Languages: Java (Expert), Python, C++, SQL

Backend Frameworks: Spring Boot, Node.js, REST APIs, GraphQL

Messaging/Streaming: Apache Kafka, RabbitMQ

Cloud & DevOps: AWS (ECS, RDS, EC2), Docker, Kubernetes, Jenkins, Git

PROJECTS

Distributed Transaction System

- Built a distributed payment processing system using Java and Kafka for asynchronous messaging.
- Implemented the Saga pattern for transaction reliability and eventual consistency.